

Exercise 5 for '*Computational Physics - Material Science*', SoSe 2025
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Please provide a well documented submission of your solution. Your submission should include

- A pdf file containing the solution to the questions with the corresponding equations that are implemented in your codes. Figures must contain axis titles with corresponding units and a caption.
- The source code should be commented, and the equations given in the pdf file have to be referenced in the source code.
- There is no need to provide the trajectory files.
- In case your code is not working properly, please provide a description of the debugging attempts you did.

Exercise: LJ fluid in the canonical ensemble (NVT)

From a statistical mechanics perspective, the temperature of an ensemble of atoms is imposed by coupling the system with a large heat bath of temperature T_0 . The velocities of the atoms in the system are then following a Maxwell-Boltzmann distribution corresponding to T_0 . The system's temperature fluctuates around T_0 . In atomistic modeling, the temperature is controlled by applying a thermostat.

Thermostatting can be achieved either by smoothly rescaling the particle velocities toward the target temperature T_0 , as in the Berendsen thermostat, or by introducing stochastic collisions between particles and an external heat bath with fixed temperature T_0 , as done in the Andersen thermostat.

- When considering a Berendsen thermostat, the velocities are scaled at each step Δt , such that the rate of change of temperature is proportional to the difference in temperature:

$$\frac{dT(t)}{dt} = \frac{1}{\tau}(T_0 - T(t)) \quad (1)$$

where τ is the coupling parameter which determines how tightly the bath and the system are coupled together. As in the simple velocity-rescaling algorithm, the velocities are thus rescaled by λ according to

$$\lambda^2 = 1 + \frac{\Delta t}{\tau} \left(\frac{T_0}{T(t)} - 1 \right), \quad (2)$$

where Δt is the simulation time step.

- When considering the Andersen thermostat, the system exchanges energy with a heat bath through stochastic collisions that randomize particle velocities. At each time step Δt , each particle has a probability $\nu \Delta t$ of undergoing a collision with the bath, where ν is the collision frequency and Δt is the time step. When a collision occurs, the velocity of

the affected particle is reassigned by sampling from the Maxwell-Boltzmann distribution corresponding to the target temperature T_0 :

$$P(v_\alpha) = \sqrt{\frac{m}{2\pi k_B T_0}} \exp\left(-\frac{mv_\alpha^2}{2k_B T_0}\right) \text{ with } \alpha = x, y, z \quad (3)$$

This stochastic process mimics interactions with a surrounding thermal reservoir. The Andersen thermostat samples the canonical (NVT) ensemble correctly in terms of equilibrium distributions, unlike the Berendsen thermostat, which does not.

The objective of this exercise is to enrich the NVE-MD engine implemented in exercise sheet #3 with different thermostat schemes (Berendsen and Andersen) to simulate a LJ fluid in a canonical (NVT) ensemble (or pseudo-canonical ensemble when considering the Berendsen thermostat). Once implemented and validated, the heat capacity of the LJ fluid is calculated as an application of the thermostat. [Completing 50% of the exercise sheet can be achieved either by implementing, validating, and comparing the two thermostat schemes (questions a) to e)), or by implementing and validating one thermostat and using it to calculate the heat capacity (question f).]

- a) Enrich the 3D MD implementation of Exercise sheet #3 with a Berendsen thermostat that rescales the velocities according to Eq. (2). Validate your implementation by performing a temperature jump experiment: To this end, simulate, during 80000 time steps, the trajectory of a system containing $N = 10^3$ atoms included in a simulation box of volume V such that the number density is $\rho = N/V = 0.5\sigma^{-3}$. The atoms are interacting through a LJ potential with $\epsilon = 0.5k_B T_0$, and use $\tau = 5000\Delta t$. Before and after the temperature jump, $T_0 = 300K$ and $T_0 = 100K$, respectively. The temperature jump is done after 40000 time steps. Plot the result $T(t)$ versus t . How does the temperature behave in the second half of the trajectory? Explain what would happen for $\tau \rightarrow \infty$ and $\tau = \Delta t$?
- b) Enrich the 3D MD implementation of Exercise sheet #3 with an Andersen thermostat that reassigned particle velocities according to Eq. (3) while considering the collision frequency $\nu = 0.01 \text{ fs}^{-1}$, and repeat the same temperature jump calculation as in a). Check and explain what would happen for $\nu \rightarrow 0$ and $\nu = 1 \text{ fs}^{-1}$.
- c) Simulate the trajectory of a system containing $N = 10^3$ atoms included in a simulation box of volume V such that the number density is $\rho = N/V = 0.5\sigma^{-3}$ during 80000 time steps using the Berendsen thermostat. The desired temperature is $T_0 = 300K$, and $\tau = 500\Delta t$. The atoms are interacting through a LJ potential with $\epsilon = 0.5k_B T_0$. The first 40000 time steps are used for the equilibration of the system, and the last 40000 time steps are for the production run. During the production run, calculate and report $\langle T \rangle$, $\langle E \rangle$, and $\langle \sigma_E^2 \rangle$.
- d) Calculate $\langle E \rangle$ also from the integration over the RDF $g(r)$, as shown in the lecture and given in the script. Are the values within the errors of the simple average?
- e) Repeat the same calculations as in c) and d) but now considering the Andersen thermostat with $\nu = 0.01 \text{ fs}^{-1}$. How do $\langle T \rangle$, $\langle E \rangle$, and $\langle \sigma_E^2 \rangle$ obtained with the Andersen thermostat compare with the ones obtained in c) and d)?
- f) The heat capacity at constant volume, C_V , can be written in the NVT ensemble as:

$$C_V = \left(\frac{\partial \langle E \rangle}{\partial T} \right)_V = \frac{\sigma_E^2}{k_B T^2} = \frac{1}{k_B T^2} \left[\sigma_U^2 + \frac{3}{2} N (k_B T)^2 \right].$$

where $\sigma_E^2 = \langle E^2 \rangle - \langle E \rangle^2$ is the variance of the total energy fluctuations, and $\sigma_U^2 = \langle U^2 \rangle - \langle U \rangle^2$ is the variance of the potential energy fluctuations.

Perform a series of MD calculations as in c) or e) but now with $T_0 = T_{init} + k\Delta T_0$ with $\Delta T_0 = 25K$, $T_{init} = 300K$ and $k \in \{-1, 0, 1\}$. For T_0 corresponding to $k = 0$, calculate and report C_V using (i) σ_E^2 and (ii) $\left(\frac{\partial \langle E \rangle}{\partial T}\right)_V$ (approximated through $dE(T_0)/dT \simeq (E(T_{init} + (k-1)\Delta T_0) - E(T_{init} + (k+1)\Delta T_0))/2\Delta T_0$). Interpret your result. Are the values comparable? If not, what could be the possible sources of error?

Numerical values of quantities to be used:

quantity	value (units)
k_B	0.0019849421 (kcal/mol/K)
ϵ	0.297741315 (kcal/mol)
σ	0.188 (nm)
T_0	300 (K)
mass	39.95 (g/mol)
Δt	1 (fs)
t_{eq}	0.04 (ns)
t_{prod}	0.04 (ns)